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Built-in Functions in Python: Study Notes

Definition:
Built-in functions are pre-defined functions in Python that are available for use without importing any external modules. They are already embedded in the Python language, which means you can use them directly in your code.

Explanation:
Built-in functions are functions that are already available in Python, making it easier for developers to use them without having to write their own code. They can perform various tasks, such as string manipulation, file handling, and mathematical operations, among others.

Advantages:
1. **Convenience**: Built-in functions are easily accessible and can save you time and effort when writing code.
2. **Efficiency**: They are optimized for performance, making your code run faster.
3. **Flexibility**: You can use them in a variety of situations and with different data types.

Disadvantages:
1. **Limited functionality**: Compared to external libraries, built-in functions have limited functionality.
2. **No flexibility in naming**: You can't change the names of built-in functions.

Examples:
1. **Print() function**: prints output to the screen
- Example: `print("Hello World!")`

2. **Len() function**: returns the length of a string or a list
- Example: `print(len("Hello World!"))`

3. **Range() function**: generates a sequence of numbers
- Example: `print(range(1, 11))`

4. **Max() and Min() functions**: return the maximum or minimum value in a list or a set
- Example: `print(max([1, 2, 3, 4, 5]))`

Important Points:

1. **Use built-in functions instead of creating your own**: For simple tasks, built-in functions are usually faster and more efficient.
2. **Study built-in functions**: Familiarize yourself with the available built-in functions to improve your programming skills.
3. **Understand the parameters**: Learn about the parameters each built-in function takes and how to use them.

Some other important built-in functions in Python include:

- `type()`: returns the type of a variable
- `range()`: generates a sequence of numbers
- `pow()`: calculates the power of a number
- `abs()`: returns the absolute value of a number
- `len()`: returns the length of a string or a list
- `min()`: returns the minimum value in a list or a set
- `max()`: returns the maximum value in a list or a set